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Simultaneous Optimization of Several Response Variables

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A problem facing the product development community is the selection of a set of conditions which will result in a product with a desirable combination of properties. This essentially is a problem involving the simultaneous optimization of several response variables (the desirable combination of properties) which depend upon a number of independent variables or sets of conditions. Harrington, among others, has addressed this problem and has presented a desirability function approach. This paper will modify his approach and illustrate how several response variables can be transformed into a desirability function, which can be optimized by univariate techniques. Its usage will be illustrated in the development of a rubber compound for tire treads.

Introduction

A common problem in product development involves the selection of a set of conditions, the X 's, which will result in a produce with a desirable combination of properties, the Y 's. Essentially, this becomes a problem in the simultaneous optimization of the Y 's, or response variables, each of which depends upon a set of independent variables, $X_1, X_2, \dots, X_p$. As an example from the rubber industry, consider the problem of a tire tread compound. Here we have a number of response variables, such as PICO Abrasion Index, 200 percent modulus, elongation at break, and hardness. Each of these response variables depends upon the ingredient variables, the X 's, such as hydrated silica level, silane coupling level, and sulfur level. We wish to select the levels for the X 's which will maximize the Y 's. Unfortunately, levels of the X 's which maximize Y_1 might not even come close to maximizing Y_2 .

One approach to this problem has been through the use of linear programming. Hartmann and Beaumont [3] and Nicholson and Pullen [5] described optimization schemes based upon the linear

programming model. However, a major disadvantage of these schemes is the philosophy upon which they are based. These methods involve optimization of one response variable subject to constraints on the remaining response variables. Often, however, the goal is the attainment of the best balance among several different response variables. In developing a compound for radiator hose, for example, it is more realistic to give water absorption, heat resistance, and tensile strength equal weights in the optimization than to optimize tensile strength while keeping the other properties within specified limits.

Harrington [2] presented an optimization scheme utilizing what he termed the desirability function. Gatza and McMillan [1] gave a slight modification of Harrington's function. We will employ a different form of this function and illustrate its use in the example of the development of a rubber compound for tire treads. In maximizing this function we will use a pattern search method similar to that presented by Hooke and Jeeves [4]. In addition, we will also plot this desirability function against two independent variables with the third held at its optimum level.

Development

Suppose each of the k response variables is related to the p independent variables by

$$Y_{ij} = f_i(X_1, X_2, \dots, X_p) + \epsilon_{ij} \quad \begin{array}{l} i = 1, 2, \dots, k \\ j = 1, 2, \dots, n_i \end{array}$$

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where f_i denotes the functional relationship between Y_i and $X_1, X_2, \dots, X_p$. We note that this function may differ for each Y_i and that f_i represents this relationship except for an error term ϵ_{ij} . If we make the usual assumption that $E(\epsilon_{ij}) = 0$ for each i , then we can relate the average or expected responses η_i to the p independent variables by

$$\eta_i = f_i(X_1, X_2, \dots, X_p) \quad i = 1, 2, \dots, k.$$

In practice, f_i typically is unknown. The usual procedure is to approximate f_i , often (but not necessarily) by a polynomial function. We then estimate η_i by $\hat{Y}_i$, the estimator obtained through regression techniques.

The desirability function involves transformation of each estimated response variable $\hat{Y}_i$ to a desirability value d_i , where $0 \leq d_i \leq 1$. The value of d_i increases as the "desirability" of the corresponding response increases. The individual desirabilities are then combined using the geometric mean.

$$D = (d_1 \times d_2 \times \dots \times d_k)^{1/k} \quad (1)$$

This single value of D gives the overall assessment of the desirability of the combined response levels. Clearly the range of D will fall in the interval $[0, 1]$ and D will increase as the balance of the properties becomes more favorable. D also has the property that if any $d_i = 0$ (that is, if one of the response variables is unacceptable) then $D = 0$ (that is, the overall product is unacceptable). It is for these reasons that the geometric mean, rather than some other function of the d_i 's such as the arithmetic mean, was used.

One-Sided Transformations

In transforming $\hat{Y}_i$ to d_i two cases arise: one-sided and two-sided desirability transformations. For the one-sided case, d_i increases as $\hat{Y}_i$ increases and is employed when $\hat{Y}_i$ is to be maximized. (Minimization of $\hat{Y}_i$ is equivalent to maximization of $-\hat{Y}_i$.) Many transformations are possible—we shall consider the transformations given by

$$d_i = \begin{cases} 0 & \hat{Y}_i \leq Y_{i*} \\ \left[\frac{\hat{Y}_i - Y_{i*}}{Y_i^* - Y_{i*}} \right]^r & Y_{i*} < \hat{Y}_i < Y_i^* \\ 1 & \hat{Y}_i \geq Y_i^* \end{cases} \quad (2)$$

and graphed in Figure 1.

The value Y_{i*} gives the minimum acceptable value of $\hat{Y}_i$. The user specifies this value of Y_{i*} , knowing that any lower value of $\hat{Y}_i$ would result in an overall unacceptable product, since $\hat{Y}_i \leq Y_{i*}$.

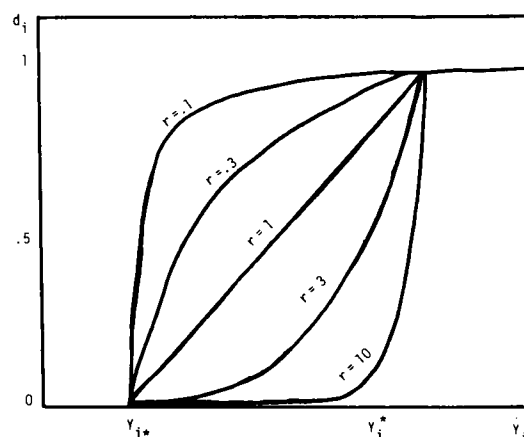


FIGURE 1. Graph of Transformation (2) for Various Values of r

would make $d_i = 0$, and thus $D = 0$, which indicates an unacceptable product. For example, if Y_i is the tensile strength of a radiator hose, a value of Y_i below $Y_{i*} = 1500$ psi would result in a product that could be unacceptable in the judgment of the management regardless of how desirable the other response variables might be.

The value Y_i^* gives the highest value of $\hat{Y}_i$. Actually, since we are considering a one-sided transformation here, there is no highest value of $\hat{Y}_i$. However, from a practical viewpoint, one can think of Y_i^* as the value for $\hat{Y}_i$ such that higher values of $\hat{Y}_i$ have little additional merit. For example, Y_i^* might be the tensile strength such that higher values of tensile strength would add little to the quality of the hose. Therefore, d_i would remain at 1.

The value of r used in the transformation would again be specified by the user. Figure 1 indicates a large value of r would be specified if it were very desirable for the value of $\hat{Y}_i$ to increase rapidly above Y_{i*} . In other words, even though Y_{i*} is an acceptable value the desirability of the product would be greatly increased by having $\hat{Y}_i$ considerably greater than Y_{i*} . Again using the radiator hose example, even though any tensile strength above $Y_{i*} = 1500$ psi would be acceptable, management might find values considerably above 1500 psi highly desirable and so choose a large value of r , say $r = 10$. As can be seen, the desirability d_i then increases slowly as $\hat{Y}_i$ increases. Therefore, to maximize d_i and thereby D , $\hat{Y}_i$ must be greatly increased over Y_{i*} . On the other hand, a small value of r would be specified if having values of $\hat{Y}_i$ considerably above Y_{i*} were not of critical importance. A value of $r = 0.1$, for example, would mean that any value of $\hat{Y}_i$ above Y_{i*} was just about as desirable as any other value of $\hat{Y}_i$ above Y_{i*} .

Two-Sided Transformations

The two-sided transformation arises when the response variable Y_i has both a minimum and a maximum constraint. We shall consider the transformations given by

$$d_i = \begin{cases} \left[\frac{\hat{Y}_i - Y_{i*}}{c_i - Y_{i*}} \right]^s & Y_{i*} \leq \hat{Y}_i \leq c_i \\ \left[\frac{\hat{Y}_i - Y_i^*}{c_i - Y_i^*} \right]^t & c_i < \hat{Y}_i \leq Y_i^* \\ 0 & \hat{Y}_i < Y_{i*} \text{ or } \hat{Y}_i > Y_i^* \end{cases} \quad (3)$$

In this situation Y_{i*} is the minimum acceptable value of $\hat{Y}_i$ and Y_i^* is the maximum acceptable value. Values of $\hat{Y}_i$ outside these limits would make the entire product unacceptable. The value selected for c_i would be that value of $\hat{Y}_i$ which was most desirable and could be selected anywhere between Y_{i*} and Y_i^* . The values of s and t in the two-sided transformations play much the same role as r does in the one-sided transformation.

In Figure 2 several different values of t and s are plotted. For illustration, it may be noted that c_i was chosen to lie at 0.25 of the distance between Y_{i*} and Y_i^* . This figure also shows that large values for s and t would be selected if it were very desirable for the value of $\hat{Y}_i$ to be close to c_i . In this case the desirability d_i would not get large until $\hat{Y}_i$ got close to c_i . On the other hand, if almost any value of $\hat{Y}_i$ above Y_{i*} and below Y_i^* were acceptable, then small values of s and t would be chosen. Moderate values for s and t (near 1) would represent a compromise between the two extremes. One could also select a large value of s and a small value of t if it were desirable for $\hat{Y}_i$ to increase rapidly to c_i while almost any value of $\hat{Y}_i$ above c_i but below Y_i^* was also desirable.

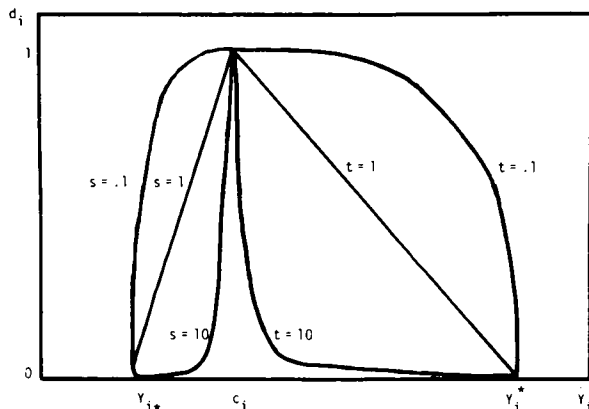


FIGURE 2. Graph of Transformation (3) for Various Values of s and t

The procedure outlined can be used to maximize some of the d_i 's (corresponding to certain $\hat{Y}_i$'s) while in essence putting constraints on the other $\hat{Y}_i$'s. This, of course, would be similar to a linear programming approach. For those $\hat{Y}_i$'s that are subject to constraints one uses extremely small values of the exponents (r , s , and t) and permits Y_{i*} and Y_i^* to act as the boundary values.

The original transformation proposed by Harrington [2] is of the form $d_i = \exp(-\exp(-Y_i))$ for the one-sided transformation and $d_i = \exp(-|Y_i|)$ for the two-sided transformation. Gatza and McMillan [1] used $d_i = \{\exp[-\exp(-Y_i)] - \exp(-1)\} / [1 - \exp(-1)]$, a modification of Harrington's which produces negative values of d_i for unacceptable properties. The transformations presented in this paper may be viewed as a type of generalization of those above. We no longer restrict ourselves to particular members of the exponential family but consider transformations that offer the user greater flexibility in the setting of desirabilities. As an example, the use of c_i in (3) allows the user to set the most desirable value of Y_i anywhere between the lower and upper boundaries (Y_{i*} and Y_i^*) rather than exactly in the middle. Harrington's and Gatza and McMillan's transformations may be closely approximated by selection of the parameters (r , s , and t) in (2) and (3) and may be viewed as special cases.

Method of Optimization

We have assumed that $\hat{Y}_i$ is a continuous function of the X_h . From (2) and (3) we see that the d_i 's are a continuous function of Y_i 's and from (1) that D is a continuous function of the d_i 's. Therefore, it follows that D is a continuous function of the X_h . As a result, existing univariate search techniques can be used to maximize D over the independent variable domain. In essence, the desirability function condenses a multivariate optimization problem into a univariate one. An added benefit of the method is the ability to plot D as a function of one or more independent variables.

Example

In the development of a tire tread compound, the optimal combination of three ingredient (independent) variables—hydrated silica level X_1 , silane coupling agent level X_2 , and sulfur level X_3 —was sought. The properties to be optimized and constraint levels were as follows.

PICO Abrasion Index, Y_1	$120 < Y_1$
200% Modulus, Y_2	$1000 < Y_2$
Elongation at Break, Y_3	$400 < Y_3 < 600$
Hardness, Y_4	$60 < Y_4 < 75$

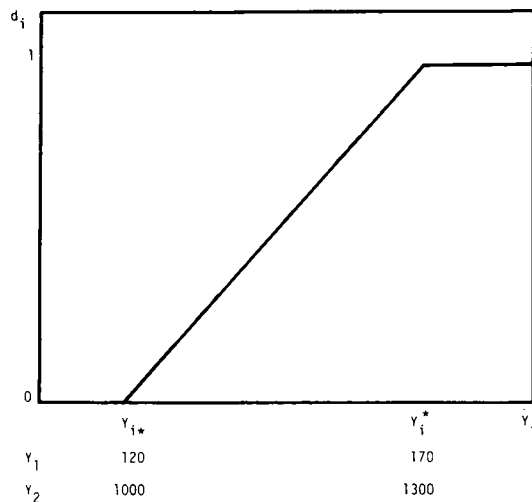


FIGURE 3. Graph of Transformation (2) used for Y_1 and Y_2 for Tire Tread Example

For Y_1 and Y_2 the one-sided transformations given by (2) were used and are shown in Figure 3. As can be seen, we set $Y_{1*} = 120$ and $Y_{2*} = 1000$. Any $\hat{Y}_i$ value less than 120 resulted in an unacceptable tire tread compound. From a practical standpoint, we set $Y_1^* = 170$ and $Y_2^* = 1300$. That is, we considered any PICO Abrasion Index above 170 to be only as desirable as one at 170. For this example, we set $r = 1$ in the transformation given by (2) for both Y_1 and Y_2 . This was done because we felt that the desirability increased in a linear manner.

For Y_3 and Y_4 the two-sided transformations given by (3) were used and are shown in Figure 4. Here $Y_{3*} = 400$ and $Y_3^* = 600$ while $Y_{4*} = 60$ and $Y_4^* = 75$. For each of these, we selected midpoints $c_3 = 500$ and $c_4 = 67.5$ as the most desirable values

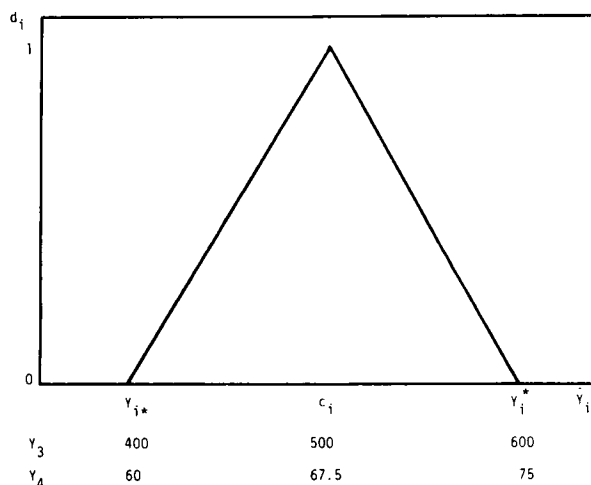


FIGURE 4. Graph of Transformation (3) used for Y_3 and Y_4 for Tire Tread Example

of Y_3 and Y_4 . Again constants of $s = 1$ and $t = 1$ were used, since we felt that a linear transformation expressed our evaluation of the desirability.

A three-variable, rotatable, central composite design with six center points (shown in Table 1) was employed to generate the data which was then fitted to the second degree polynomials

$$\hat{Y}_i = b_0 + \sum_{L=1}^3 b_L x_L + \sum_{L=1}^3 \sum_{m=L}^3 b_{Lm} x_L x_m \quad (4)$$

$$i = 1, 2, 3, 4.$$

The resultant fitted coefficients are given in Table 2, along with the standard errors for each Y_i . Since it is important to have a good estimator $\hat{Y}_i$ of η_i for this optimization technique care should be taken to use good regression and design techniques, along with experience. It was felt from past experience that at least a second degree polynomial would be required to provide an adequate fit to the data. A central composite response surface design was employed because of favorable past experience with such designs. With less previous experience, however, one could certainly utilize standard procedures in design and regression (including stepwise regression) in obtaining estimators $\hat{Y}_i$.

The next step was to use the coefficients given in Table 2 along with various values of X_1 , X_2 , and X_3 to obtain the $\hat{Y}_i$'s. Each $\hat{Y}_i$ was then transformed into a d_i using (2) and (3) as illustrated in Figures 3 and 4. The four d_i 's were combined into a single D using (1). Hence, for each level of X_1 , X_2 , and X_3 , a D value was obtained. We then searched through

TABLE 1. Experimental Design

Compound No.	x_1	x_2	x_3	Y_1	Y_2	Y_3	Y_4
1	-1	-1	+1	102	900	470	67.5
2	+1	-1	-1	120	860	410	65
3	-1	+1	-1	117	800	570	77.5
4	+1	+1	+1	198	2294	240	74.5
5	-1	-1	-1	103	490	640	62.5
6	+1	-1	+1	132	1289	270	67
7	-1	+1	+1	132	1270	410	78
8	+1	+1	-1	139	1090	380	70
9	-1.633	0	0	102	770	590	76
10	+1.633	0	0	154	1690	260	70
11	0	-1.633	0	96	700	520	63
12	0	+1.633	0	163	1540	380	75
13	0	0	-1.633	116	2184	520	65
14	0	0	+1.633	153	1784	290	71
15	0	0	0	133	1300	380	70
16	0	0	0	133	1300	380	68.5
17	0	0	0	140	1145	430	68
18	0	0	0	142	1090	430	68
19	0	0	0	145	1260	390	69
20	0	0	0	142	1344	390	70

$$x_1 = (\text{phr silica} - 1.2)/0.5$$

$$x_2 = (\text{phr silane} - 50)/10$$

$$x_3 = (\text{phr sulfur} - 2.3)/0.5$$

where x_1 , x_2 , and x_3 are design levels

(phr = parts per hundred)

TABLE 2. Regression Coefficients and Standard Error for Responses

	b_0	b_1	b_2	b_3	b_{11}	b_{22}	b_{33}	b_{12}	b_{13}	b_{23}	Std. Error
(Y_1)	139.12	16.49	17.88	10.91	-4.01	-3.45	-1.57	5.13	7.13	7.88	5.61
(Y_2)	1261.11	268.15	246.50	139.48	-83.55	-124.79	199.17	69.38	94.13	104.38	328.69
(Y_3)	400.38	-99.67	-31.40	-73.92	7.93	17.31	0.43	8.75	6.25	1.25	20.55
(Y_4)	68.91	-1.41	4.32	1.63	1.56	0.06	-0.32	-1.63	0.13	-0.25	1.27

the levels of X_1 , X_2 , and X_3 to find the optimum value for D . All of this was, of course, done on a computer. The algorithm we employed generally converged in fewer than 250 iterations. The resulting optimum formulation is shown in Table 3. The maximum composite desirability was 0.583 and all of the constraints have clearly been met. The value of 0.583 has little numerical meaning, except to indicate the level of the X 's where the maximum D occurs. Aside from finding the maximum D , however, the experimenter is generally interested in how stable the optimum is. For example, do small changes in the independent variables result in sharp decreases in D ? Since D is a function of the X variables, it can be plotted to answer such questions.

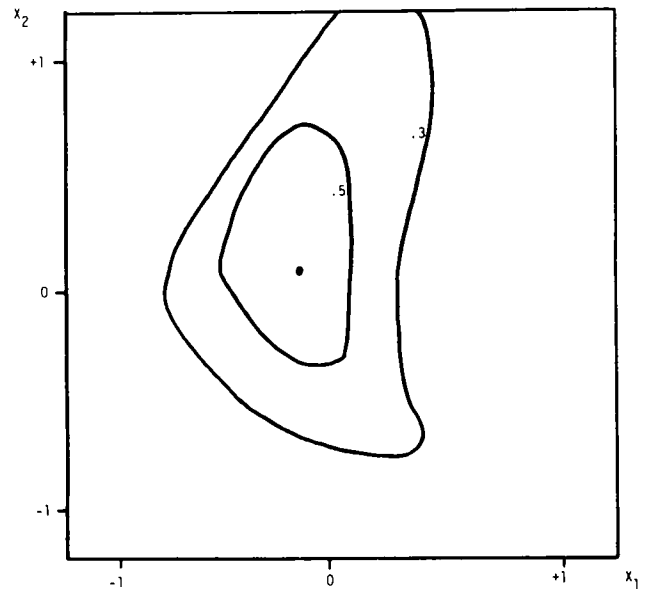
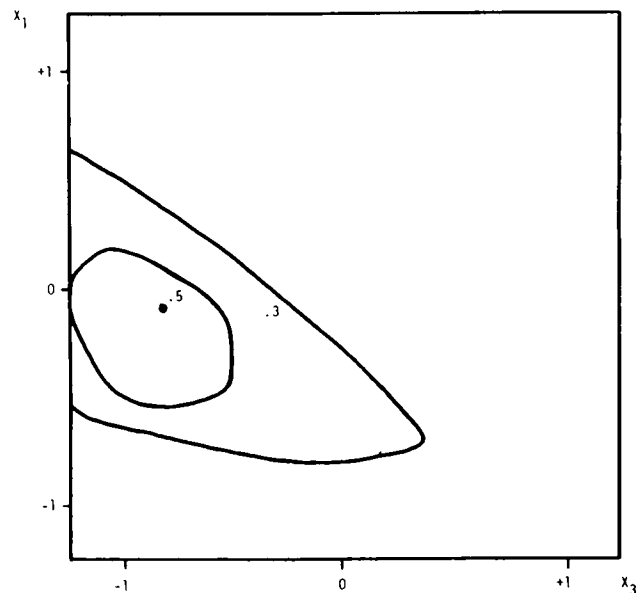
Figures 5, 6, and 7 show the contour plots (sketched from a grid of D values) of D for two independent variables with the other held at its

TABLE 3. Optimum Compound and Predicted Properties

X_1 = -0.050	
X_2 = 0.145	
X_3 = -0.868	
Y_1 (PICO) = 129.5	d_1 = 0.189
Y_2 (Modulus) = 1300	d_2 = 1.000
Y_3 (Elongation) = 465.7	d_3 = 0.656
Y_4 (Hardness) = 68.0	d_4 = 0.932

Maximum composite desirability, $\hat{D}$ = 0.583

Y_{1*} = 120	
Y_{2*} = 1000	
Y_{3*} = 400	Y_{3*} = 600
Y_{4*} = 60	Y_{4*} = 75

FIGURE 5. Contour Plots of D for X_1 and X_2 for Tire Tread ExampleFIGURE 6. Contour Plots of D for X_1 and X_3 for Tire Tread Example

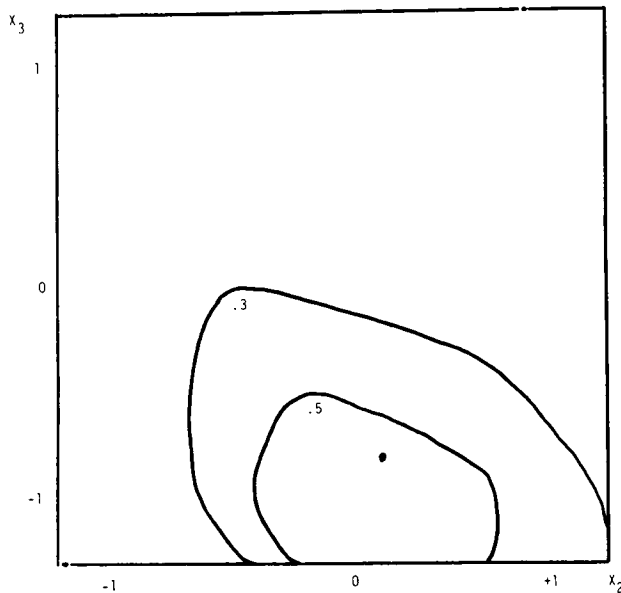


FIGURE 7. Contour Plots of D for X_2 and X_3 for Tire Tread Example

optimum. For example, Figure 5 shows the plot of X_1 versus X_2 with X_3 held at its optimum, that is, $X_3 = -0.868$. All three of these plots show the surface to be relatively flat near the maximum, meaning that small departures from optimality of the X values would not appreciably decrease the desirability.

Obviously, the approach utilized in this example is not the only possible approach. Another feasible method would involve studying the coefficients in the fitted equations and overlaying contour plots. However, the optimum reached in Table 3 did prove to be satisfactory from a production standpoint, although slight deviations from the optimum levels of the X 's were instituted for other reasons. This proved no great problem in this example, since the surface is relatively flat near the optimum.

Computer Program

We have available, and will provide upon request, a copy of the FORTRAN computer program used to maximize D in terms of the X_h . This program also enables one to generate a response surface of D as a function of two of the independent variables, holding the other independent variables constant. This can then be used to obtain contour plots. It should be noted that any good optimization program may be used.

Summary

The simultaneous optimization of several responses has often been accomplished by a hit-or-miss approach. In such a procedure, numerous formulations are evaluated until one is found which is within all constraints. This becomes the "optimum" formulation. The desirability function approach is a considerable improvement over this method and usually not only requires fewer formulations to be evaluated but also results in more desirable property levels. Furthermore, the advantage of being able to plot the desirability surface to determine its sensitivity to small changes in the independent variables is significant.

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